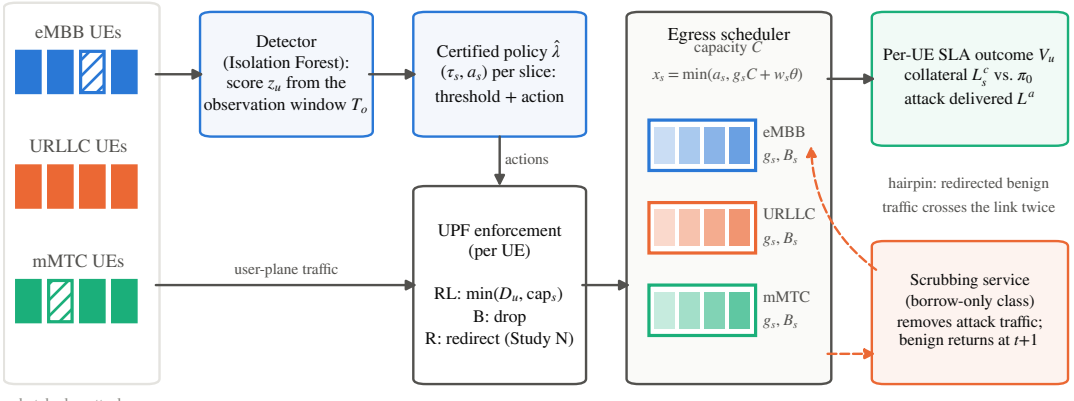
eMBB UEs
URLLC UEs
mMTC UEs
hatched = attacker
Detector
(Isolation Forest):
score z_u from the
observation window T_o
Certified policy $\hat{\lambda}$
(τ_s, a_s) per slice:
threshold + action
actions
user-plane traffic
UPF enforcement
(per UE)
RL: $\min(D_u, \text{cap}_s)$
B: drop
R: redirect (Study N)
Egress scheduler
capacity C
$x_s = \min(a_s, g_s C + w_s \theta)$
eMBB
g_s, B_s
URLLC
g_s, B_s
mMTC
g_s, B_s
Per-UE SLA outcome V_u
collateral L_s^c vs. π_0
attack delivered L^a
hairpin: redirected benign
traffic crosses the link twice
Scrubbing service
(borrow-only class)
removes attack traffic;
benign returns at $t+1$